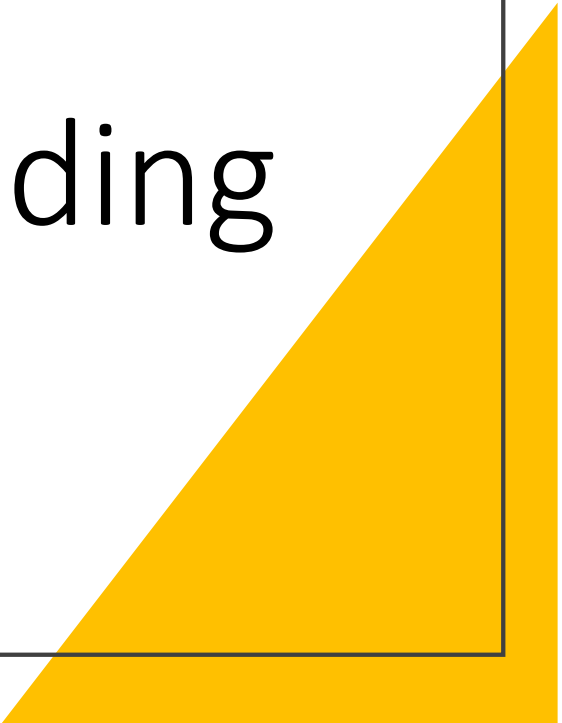


Data Analytics Technology Extraction Transformation Loading

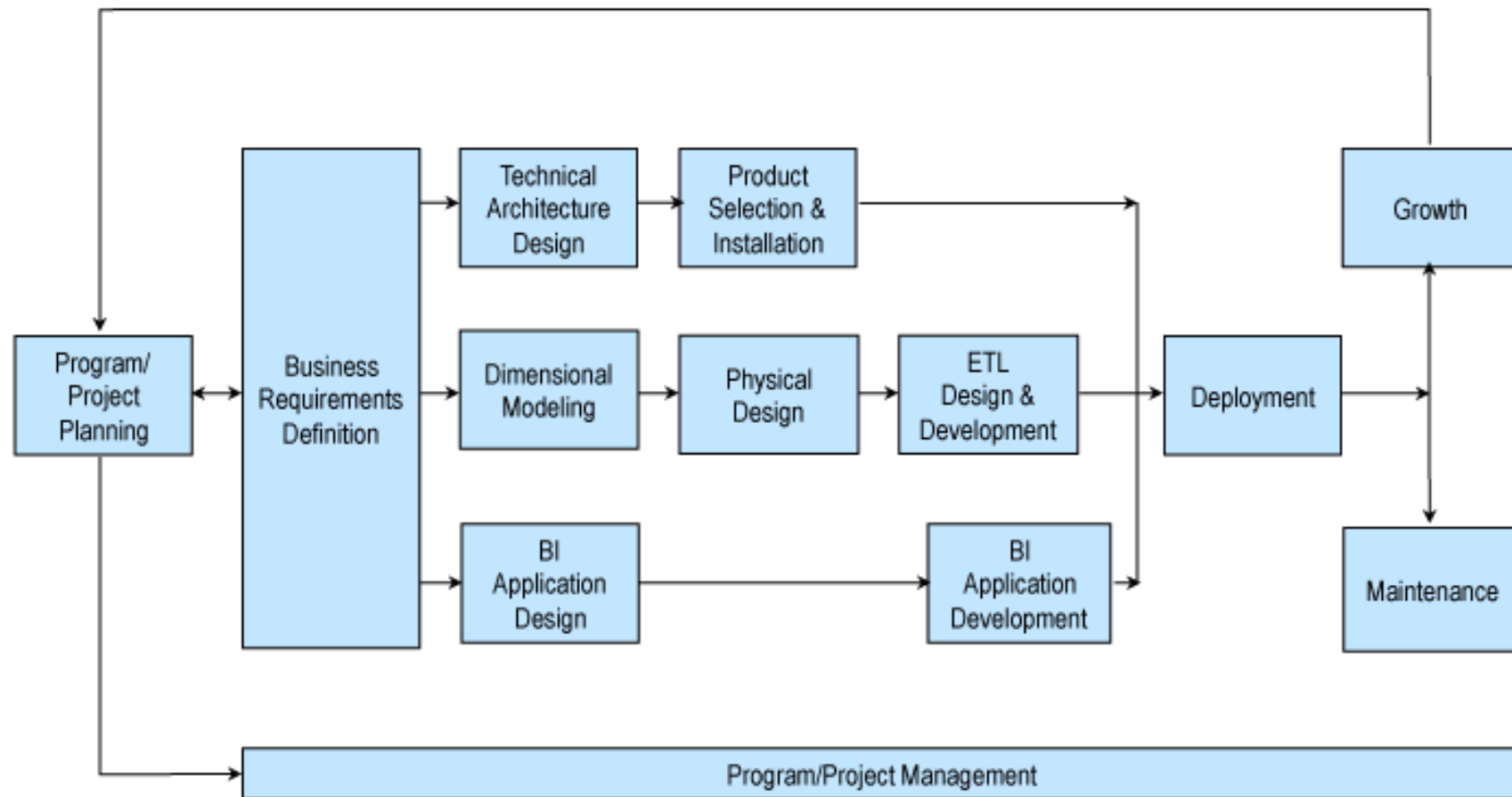
Catarina Silva

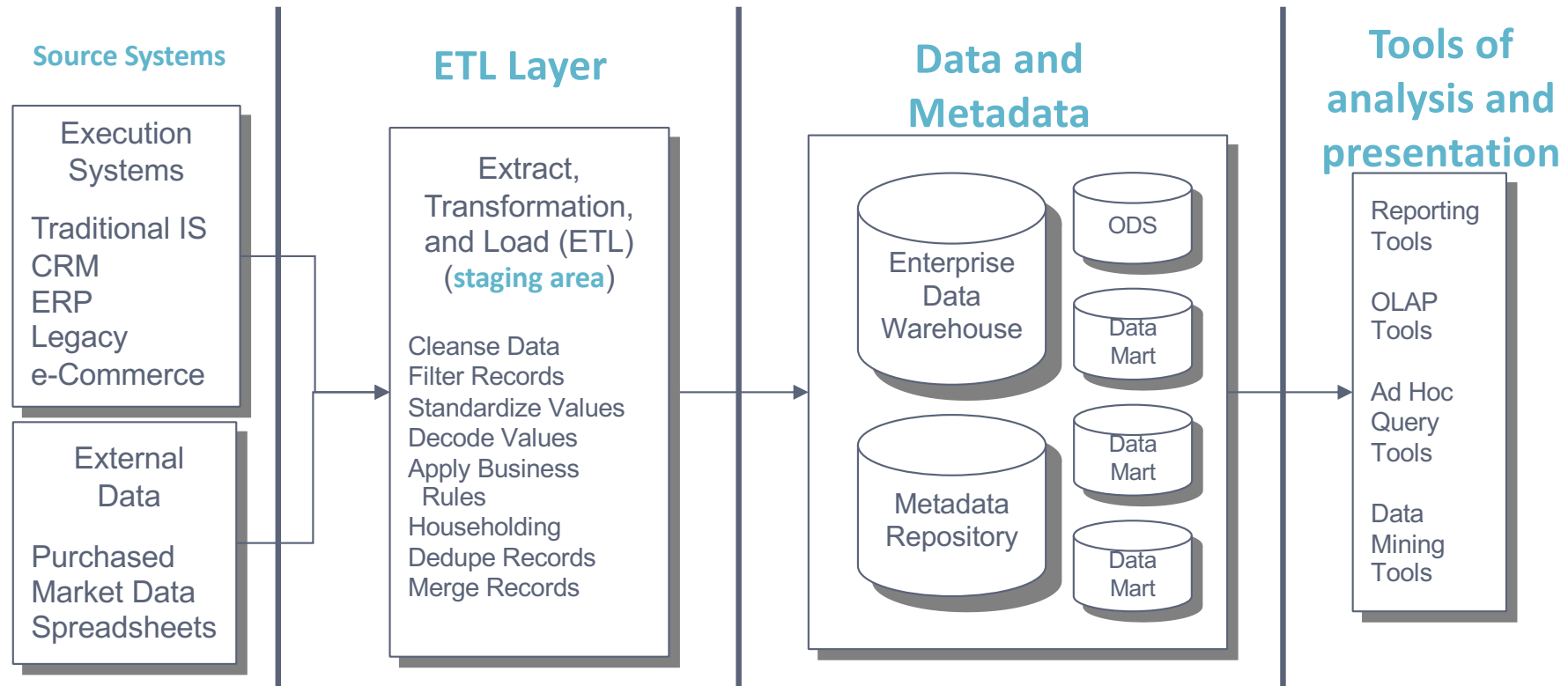
catarina@dei.uc.pt

2026

A large yellow triangle is positioned in the bottom right corner of the slide, pointing towards the top right.

Data Analytics Project





Technologies:

PeopleSoft
SAP
Siebel
Oracle Applications
Manugistics
Custom Systems

ETL Tools:
Informatica PowerMart
ETI
Oracle Warehouse Builder
Custom programs
SQL scripts

Oracle
SQL Server
Teradata
DB2

Custom Tools
HTML Reports
Cognos
Business Objects
MicroStrategy
Oracle Discoverer
Brio
Data Mining Tools
Portals

What is ETL?

- In a simplistic way, ETL ("Extract, Transform, and Load") is a way to migrate data from one database to another (the DW), making the necessary transformations in the data (format and content).
 - Transport: Possible extra step to migrate data from one location to another.
- ETL is not the mere juxtaposition of three well-defined processes:
 - there is great interdependence between them (but, from a purely pedagogical perspective, they can be discussed independently)
- ETL is (maybe) the BIG problem in DW.
 - Usually consumes more than 50% of the costs of building and maintaining a DW.

ETL variations

- E+T+L
 - Extract, Transform, Load
- E+T+T+L
 - Extract, Transport, Transform, Load
- E&T + T + T&L
 - Extract&Transform, Transport, Transform&Load

Extraction

- It needs the "cooperation" of the source systems:
 - Exporting data
 - Typically, they are converted into a file which is then read into the stage area.
 - Extraction of data
 - Using specific code that directly transports the data to the stage area.
- Two very different situations:
 - First Data Extraction
 - Incremental extractions after first extraction

Transformation



- Cleaning of data
- Data integration
- Other transformations
 - Modify codes, calculated values, previous aggregations, introduction of temporal references for exceptional cases, etc.

Transport

- This step is not always necessary
- Copy, move data to other DW (s)
- Possible technical difficulties
 - Synchronization
 - Bandwidth
 - Quantity of data to carry

Data cleanse

Clear "dirty" data:

- No-nonsense values
- Absence of data
- Duplicate data
- Data whose meaning is not clear (and that the metadata does not clarify)
- Contradictory data
- Data that violates basic integrity rules (keys, integrity, etc.)

Eliminate inconsistencies

- Due to the collection of the same data in more than one system
- Due to insufficiencies in the extraction process
- Caused by changes in operating systems
- Due to technical problems in operating systems

Example of data that require cleansing

CUST #	NAME	ADDRESS	TYPE
90328574	Digital Equipment	187 N. PARK St. Salem NH 01458	OEM
90328574	DEC	187 N. Pk. St. Salen NH 01458	
90238475	Digital	187 N. Park St Salem NH 01458	\$#%
90233479	Digital Corp	187 N. Park Ave. Salem NH 01458	Comp
90233489	Digital Consulting	15 Main Street Andover MA 02341	Consult
90234889	Digital Info Service	PO Box 9 Boston MA 02210	Mail List
90345672	Digital Integration	Park Blvd. Boston MA 04106	SYS INT

Non-unique keys

Duplicates

Absence of standardization

Mistakes

Noise and null data

Data cleanse process

Automatic processes / tools. They usually resolve effectively:

- Problems related to data formats: conversions, etc.
- Lack of standardization
- Fill in missing values
- Etc

Manual processes

- Required when the hotfix is semantic
- Supported by tools

		Antunes, Rui
Ana	Silva	
		Marques, José
Manuel	Neves	

Format



		Rui Antunes
Ana	Silva	
		José Marques
Manuel	Neves	

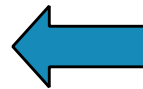


Attribute
separation

Semi-automatic cleanse

Rui		Antunes
Ana	Silva	
José		Marques
Manuel	Neves	

Fusion



		Rui	Antunes
Ana	Silva		
		José	Marques
Manuel	Neves		

Fusion



Rui	Antunes
Ana	Silva
José	Marques
Manuel	Neves

Data Integration

- Two main problems:
 1. Data that should be related but which can not be properly related
 - For example, due to the absence of primary keys in the data or to non univocal keys
 2. Data that is related but that, in fact, should not have any relationship between them
 - For example, it occurs when attributes or records are used for various purposes

Loading

- Critical phase:
 - possible failures can lead to complex recoveries (and redundancies!)
- Almost everything that is done to optimize DW performance tends to delay loading:
 - Indexes
 - Partitioning tables
 - Materialized views
 - Parallelism and distribution
 - Server Statistics
 - Logging systems
- In addition to the initial loading, it is necessary to resolve the periodic loads, which have different characteristics (e.g., updates of dimensions, etc.).
 - **It is important to minimize the loading window.**

The typical ETL process

Planning

- Define a general plan (of the end-to-end type)
- Define infrastructure for the staging area
- Choosing ETL Tools
- Make detailed plan analyzing all the problems that need to be solved to load each target table (e.g., sources, transformations, etc.)

Dimensional loading

- Make, test and execute ETL plans for static and simple dimensions. Let's you test the entire infrastructure.
- Make, test and execute ETL plans for the dimensions that change.
- Treat all other cases (generated dimensions, with manual data, etc.)

Uploading facts

- Make, test, and execute ETL plans for fact tables
- Make and test periodic loading process

Automate the process to the maximum

Product

ID_product
Number
Name
Brand
Category
Subcategory
Department
Package_size
Package_type
Diet_Type
Weight
Weight_unit
Units_box
Boxes_pallet
Shelf_lenght
Shelf_height
Shelf_depth
.....

Time

ID_date
Day_of_month
Day_of_week
Day_of_year
Week_of_year
Month
Month_number
Trimester
Tax_year
Flag_holiday
Flag_week_day
Flag_last_day_month
Season
Special_event
.....

Promotion

ID_promotion
Number
Promo_name
Type_Price_reduct
Type_advertising
Type_poster
Type_coupons
Advertise_media
Poster_media
Promotion_cost
Start_promotion
End_promotion
.....

Sale

ID_product
ID_date
ID_store
ID_promotion
ID_client
Units_sold
Aquisition_cost
Sale_price
Amount_of_clients

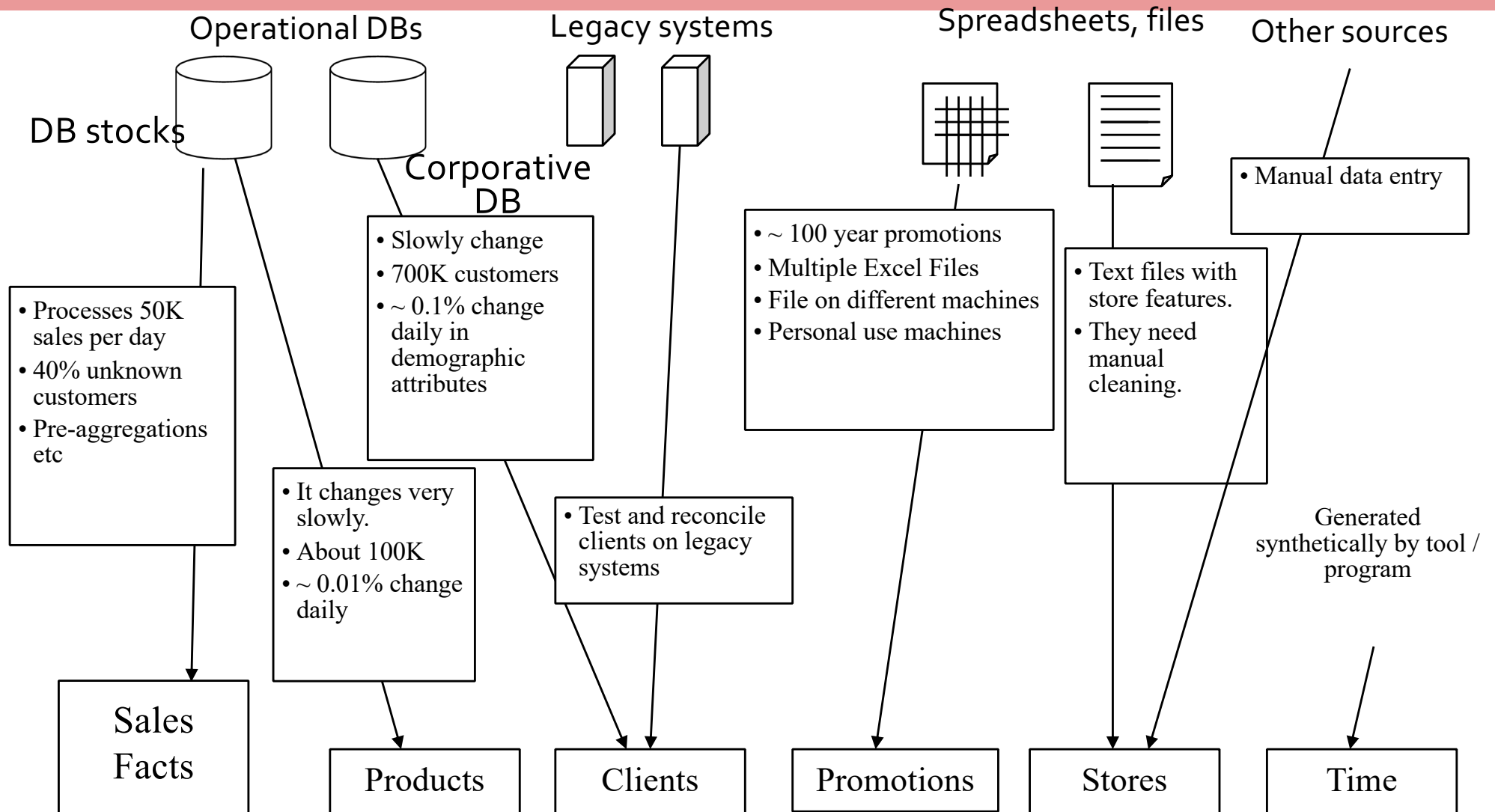
Clients

ID_client
Card_number
Name
Address
City
Zio_code
State
Region
Telephone
(Other demographic data)
.....

Store

ID_store
Name
Store_number
Address
City
Zip_code
State
Region
Telephone
Fax
Store_manager
Total_area
Grocery_area
Frost_area
Amount_boxes
Inauguration_date
Last_remodelin_date.
.....

Data Sources



Destinations

Defining the infrastructure for the staging area

- You can go from a simple account on the server where DW will be dedicated to large capacity machines.
- The decision depends on the volume of data involved and the complexity of the operations to be performed on the data before loading them into DW.
- Typically, for each dimension and fact table, everything is prepared in the staging area and bulk loaded to the DW afterwards.
- The staging area should take into account that partial loads are frequent for very large facts and dimensions.

Initial loads

- Made directly from the staging area to the DW tables (after the data prepared)
- Some things to consider:
 - Disconnect (or configure for minimal impact on performance) logging systems.
 - Pre-order the data to be loaded by the primary key
 - Do some basic aggregations eventually during the load.
- Index management:
 - Deactivate + reindex if you enter more than 10% to 15% of the records
 - Keep the indexes. In this case it is necessary to consider whether the physical structures of the indexes are prepared for growth.

Periodic loadings

- Define strategy to identify new data in source systems (a huge problem!!!):
 - New transactions
 - Updates to past transaction data
- Identify:
 - New registrations to be introduced in each dimension
 - Dimension attribute updates and how they will be treated
 - New facts

Identifying new data on the data sources

- Use the existing log system in the source system
- Use characteristics of the applications of the systems sources:
 - Use of timestamps;
 - Rules used for assigning keys
 - Attributes of certain tables that determine moments in time with precision.
 - Signatures
 - Etc.
- If possible, build a log system specific to source systems (e.g., using snapshots):
 - Interferes with source system performance.
 - Not always feasible

Research problems (1)

Extraction

- Access to source systems
- Identification of new data to load
- Minimize interference with source systems
- Dealing with changes in source systems

Transformation

- Defining Rules for Data Transformations
- Data cleansing: automate to the maximum, even when there is semantics involved
- Data quality metrics and quality assurance process
- Impact on transformation processes of changes in source systems

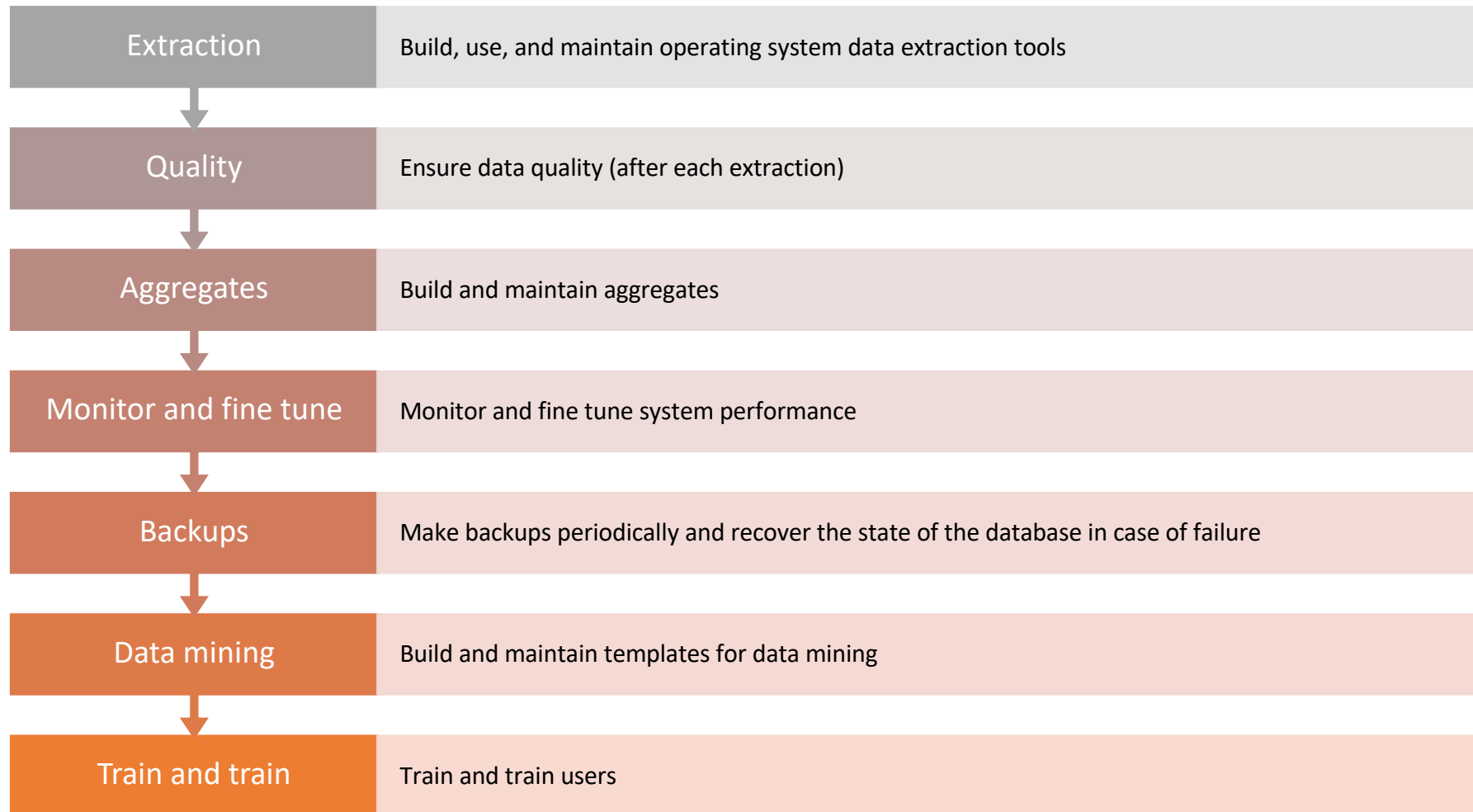
Loading

- Minimize (or reduce to zero) the loading window.

Research problems (2)

- General Process Issues
 - Ensure the quality of the final data
 - Metadata management
 - Source system metadata
 - Import
 - Maintenance, when metadata changes in source systems
 - DW target metadata
 - Metadata of the staging area, including the metadata of all transformations to be made to the data.
 - Formats for metadata, including existing and emerging standards
- ETL General Process Management
 - Centralized view
 - Audit
 - Error detection and recovery in case of failure
 - Scheduling tasks

DW administration and ETL

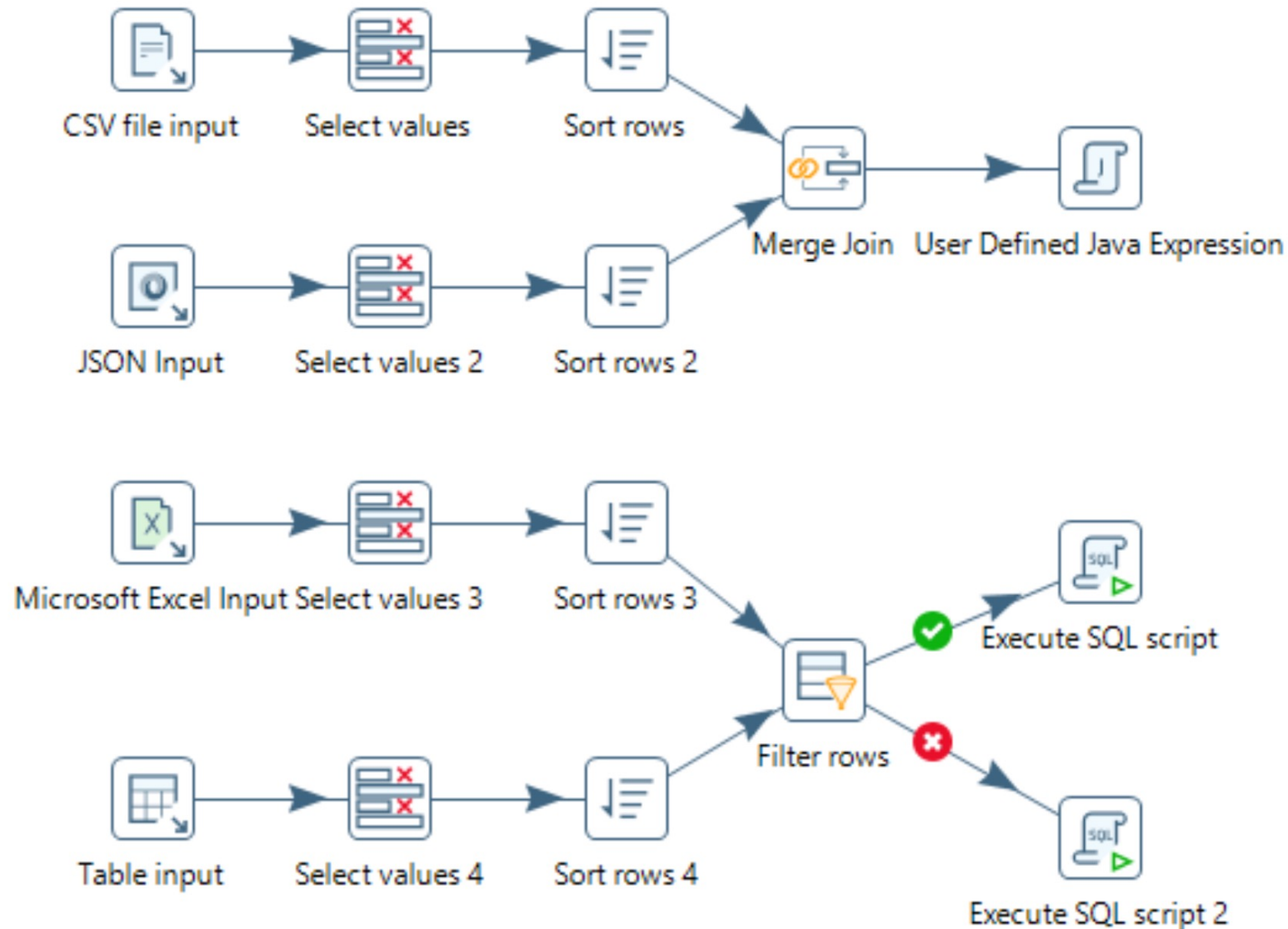




Typical rhythm of the DW and the ETL

- The DW transits periodically (daily, weekly, etc.) between two states:
 - Exploration (16 to 22 hours per day in the case of a daily pace)
 - Charging (typically 2 to 8 hours per day in the case of a daily pace, but this window needs to, and it is becoming smaller)
 - Load new data;
 - (Re) build indexes and other structures necessary to optimize performance;
 - Check data quality;
 - Open access to new data.

Example – ETL process



“Facts are obsolete — now it’s just about whether I can get away with it...”



timoe Elliott.com